MILITARY SPECIFICATION

TEST SET, COUNTERMEASURES SET

TS-3615( )/ALQ-136(V)1

This specification is approved for use by the U. S. Army Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements for Test Set, Countermeasures Set TS-3615( )/ALQ-136(V)1, referred to hereinafter as the bench test set (BTS).

1.2 Description. The BTS is a type of Special Test Equipment used to test the operation of the Countermeasures Set AN/ALQ-136(V)1, hereinafter referred to as the CMS, at the direct support level of maintenance, Aviation Intermediate Maintenance (AVIM).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents form a part of this specification to the extent specified herein. The effective issue or revision of specifications and standards shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (a) specific issues are set forth herein, or (b) different issues than those specified in this specification are set forth in the solicitation. The date of the applicable DODISS and supplements thereto, and the effective issue or revision of drawings cited herein, shall be as specified in the solicitation or contract.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAERADCOM, ATTN: DELET-DP, Fort Monmouth, N. J. 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated, Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet-stock (Container Grade) and Cut Shapes

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268K(EL)	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231(EL)	Marking of Electronic Items
MIL-F-14072(EL)	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252(EL)	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A w/Notice 4(EL)	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 w/Notice 3(EL)	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810C	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

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2.1 Issues of documents (cont'd)

DRAWINGS

U.S. ARMY ELECTRONICS RESEARCH AND DEVELOPMENT COMMAND

SM-A-919708	Letter Code (Classified)
DL-SM-B-919748	Test Set, Countermeasures Set TS-3615( )/ ALQ-136(V)1
SM-A-920091	Acceptance Test Procedure for Test Set, Countermeasures Set TS-3615( )/ALQ-136(V)1
SM-A-920096	Environmental Test Procedure for Test Set, Countermeasures Set TS-3615( )/ ALQ-136(V)1
SM-A-920182	Burn-in Procedure
SM-HL-920192	PRI Threat A-W Data (Listing)
SM-TP-920192	PRI Threat A-W Data (Punched paper tape)
SM-HL-920193	PRI Threat A-F Data (Listing)
SM-TP-920193	PRI Threat A-F Data (Punched paper tape)
SM-HL-920194	PRI Threat B Data (Listing)
SM-TP-920194	PRI Threat B Data (Punched paper tape)

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained form the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.4 (see 6.2 and 6.6).

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and drawing DL-SM-B-919748 which describes Test Set, Countermeasures Set TS-3615( )/ALQ-136(V)1, including the requirements for parts, materials and processes thereon.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the general requirements of MIL-P-11268K(EL) including the selection requirements therein, shall apply.

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3.3.1 Finish. The equipment shall be finished in accordance with MIL-F-14072(EL) and the equipment drawings.

3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231(EL) (see 4.5). Front panel marking shall be Group I as described in MIL-M-13231(EL). Location of marking shall be as specified on the equipment drawings.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial numbers. The following items shall have serial numbers:

SM-D-919748	Test Set, Countermeasures Set
SM-D-919749	Control Panel Assembly
SM-B-919754	Control Box
SM-D-919750	Status Panel Assembly
SM-D-919751	RF Modulation Assembly
SM-D-919809	Circuit Card Assembly, Pulse Generator
SM-D-919810	Circuit Card Assembly, CS and Pulse Control

3.5 System performance. The BTS tests the operation of the CMS at the Aviation Intermediate Maintenance Level (AVIM). It consists of a combination case which houses all the threat generation circuitry as well as status indication circuits and forcing functions. It utilizes GFE test equipment which is necessary for the performance of electrical tests on the CMS. The GFE test equipment is located outside the combination case.

3.5.1 Primary power. The BTS shall operate and maintain specified performance using 105-125 volt, 50-60 Hz, single phase primary input.

3.5.1.1 CMS primary power. The primary power for the CMS is supplied through the BTS from an auxiliary 28 Vdc power supply external to the BTS. (see 6.8).

3.5.2 Control panel assembly. The control panel assembly contains the following items.

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3.5.2.1 Operator control unit (OCU). The OCU of the BTS controls the operating mode of the CMS during bench testing and contains a 3 minute warm-up timer to control initial operation. It has a 3 position toggle switch for selection of OFF, STBY and OPR modes and three indicator lights corresponding to ON, STBY and OPR modes of the CMS. Required functions are:

SWITCH POSITION	INDICATOR LIGHTS "ON"	CMS STATUS
OFF	All Lights OFF	CMS turned OFF, no activity indicated.
STANDBY	ON	CMS is in warm-up mode until lapse of 180 + 20 sec from switch position OFF.
STANDBY	ON and STBY	Indicates 180 sec. delay has elapsed and CMS may be operated.
OPERATE	ON and OPR	CMS is fully operational.

3.5.2.2 Indicator lights. Two indicator lights on the control panel assembly demonstrate the operational state of the CMS and are provided with a self test feature. These lights shall indicate JAM, and INOP conditions of the CMS.

3.5.3 Status panel. The following functions are incorporated on the status panel.

3.5.3.1 Status signals. The signals listed below shall indicate the status of the system by means of panel indicator lights.

CHANNEL 1, 2	SORTER LOADED
CHANNEL 1, 2	SORTER LOAD COMMAND
CHANNEL 1, 2	A
CHANNEL 1, 2	B
CHANNEL 1, 2	SLO LOAD COMMAND
CHANNEL 1, 2	SLO LOADED
CHANNEL 1, 2	SLO ACCUR
CHANNEL 1, 2	TC RANGE JAM
CHANNEL 1, 2	CON SCAN

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3.5.3.2 Forcing functions. Forcing functions shall be a circuit ground (< 1 ohm in the BTS) on a single line to force portions of the CMS system into selected modes for system test purposes. The forcing function outputs are given below:

SORTER 1, 2 LOAD
SORTER 1, 2 OVRD
MOD DRIVE INHIBIT
NOISE INHIBIT
SW-3 INHIBIT
CP
RESET

3.5.3.3 Voltage monitor. The dc voltage outputs of the low voltage power supply utilized within the CMS shall be made available as instrumentation outputs and shall be monitored at the BTS with a digital multimeter. Each voltage shall be selected for output monitoring by a multi-position rotary switch with the following positions available:

+5V LOGIC
+5V
-5.2V
+10V
-10V
+15V
-15V
+28V
-65V

3.5.3.4 Oscilloscope monitor. The signals listed below shall be selectable by dual concentric multiposition rotary switches operating in conjunction with 2 front panel BNC connectors, CHAN 1 and CHAN 2.

SORTER 1, 2 BLANK
SORTER 1, 2 TG
SORTER 1, 2 TC
SORTER 1, 2 ING
TC1, 2 VCO
RNG SYNC 1, 2
LM DRIVE
PULSE VIDEO
LOG VIDEO
LIM VIDEO
THRESH CONT
BTS VIDEO 1/2

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3.5.3.5 Lamp test. The status panel shall be provided with lamp test capabilities. All lamps except the primary POWER lamp shall be tested. The lamp function shall be initiated by a momentary action switch.

3.5.4 RF modulation assembly. There are two identical RF modulation assemblies in the BTS. All RF output circuitry shall be capable of operating over the A to E +0.5 GHz frequency range except that the voltage controlled oscillator (VCO) frequency shall only cover D to E +0.5 GHz. Power at any output port shall not exceed +20 dBm peak or average. Step attenuators shall provide RF attenuation in 1 dB steps from 0 to 69 dB (see SM-A-919708).

3.5.4.1 RF output. The RF modulation assembly shall be capable of supplying an RF signal at a single output port. This signal shall simulate the RF input signal to the AN/ALQ-136(V)1 RF input port.

Frequency and switch positions	F1:	D + 0.25 $\pm$ 0.15 GHz
	F2:	D + 1.25 $\mp$ 0.15 GHz
	F3:	D + 2.25 $\mp$ 0.15 GHz
Max power		+5 dBm minimum

Output attenuation: adjustable in nine 1.0 $\pm$ 0.1 dB steps and six 10.0 $\pm$ 0.5 dB steps

RF selection: either pulse or CW

Impedance: 50 ohms, nominal (no test required)

3.5.4.2 Pulse characteristics. Four separate pulse modulated trains shall be generated in each RF modulation assembly one at a time. Two threats, one from each RF modulation assembly can be applied to the CMS at one time. The pulse trains shall be selected by the PRI switch having the following positions:

OFF
VAR PRI
THREAT A-W
THREAT A-F
THREAT B

3.5.4.2.1 Pulse width. The pulse width shall be 0.1 usec to 9.9 usec, adjustable by thumbwheel, in 0.1 usec steps.

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3.5.4.2.2 PRI. The VAR PRI control shall be adjustable from 300 usec to 999 usec in 1 usec steps. The PRI modulation for the A-W threat shall have PRI limits of 260 and 380 usec and shall be obtained from data existing on punched paper tape SM-TP-920192 and on listing SM-HL-920192. The PRI modulation for the A-F threat shall have PRI limits of 267 and 274 usec and shall be obtained from data existing on punched paper tape SM-TP-920193 and on listing SM-HL-920193. The PRI modulation for the B threat shall have PRI limits of 347 and 408 usec and shall be obtained from data existing on punched paper tape SM-TP-920194 and on listing SM-HL-920194.

3.5.4.2.3 Con scan modulation. The con scan (CS) modulation generator shall produce sinusoidal modulation on the RF output with a frequency range of 1 to 99 Hz. A control on the RF modulation assembly 1 and 2 front panels shall adjust the modulation depth of the pulsed RF train (defined as the dB difference between the crest and trough of the sinusoidally modualted signal) from 0 dB to 25 dB in the following steps:

Step 1	0.0 dB
2	1.0 dB $\pm$ 0.1 dB
3	2.0 dB $\mp$ 0.2 dB
4	3.0 dB $\mp$ 0.3 dB
5	6.0 dB $\mp$ 0.6 dB
6	9.0 dB $\mp$ 0.9 dB
7	12.0 dB $\mp$ 1.0 dB
8	15.0 dB $\mp$ 1.0 dB
9	20.0 dB $\mp$ 1.0 dB
10	25.0 dB $\mp$ 1.0 dB

3.5.4.3 Sync outputs.

3.5.4.3.1 Pulse sync. A TTL logic 1 signal shall be provided that shall be synchronized to the pulse modulation generated by the RF modulation assembly. The sync signal shall precede the pulse modulating signal by 0.8 $\pm$ 0.15 microseconds (leading edge to leading edge).

3.5.4.3.2 CS (pattern) sync. A TTL logic 1 signal shall be provided that shall be synchronized to the scan modulation generated by the RF modulation assembly. The sync shall be within 0.5 milliseconds of the negative going zero crossing point of the scan modulation signal.

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3.5.4.4 Blanking output. A TTL logic 0 signal shall be provided that shall precede the leading edge of each RF output pulse by 50 to 200 nanoseconds. The blanking pulse shall be 2.0 ± 0.2 microseconds wide.

3.5.5 Combined RF output. The RF output from each RF modulation assembly shall be combined and brought to an output port. The power level at this port for each RF modulation assembly shall not be less than -10 dBm.

3.5.6 RF in. The transmitted output from the CMS shall be monitored utilizing a 30 dB attenuator mounted directly on the CMS LRU-1 RF OUT connector (1J4). It shall then be connected to the RF IN connector on the status panel and the modulated RF shall be futher attenuated within the BTS by a 0-69 dB adjustable attenuator, then detected and the resultant signal made available for oscilloscope presentation. The undetected attenuated RF signal shall also be available for oscilloscope presentation.

3.6 Environmental requirements. The equipment shall meet the following service conditions without degradation in specified performance. Where a test is referenced, meeting the performance criteria shall be considered as compliance with the requirements.

3.6.1 Temperature.

3.6.1.1 Low temperature. The BTS shall be capable of operation at ambient temperatures as low as +32°F and of being stored at temperatures as low as -70°F. Before, during and after testing as specified in 4.9.1.1 the BTS shall meet the performance specified in SM-A-920096.

3.6.1.2 High temperature. The BTS shall be capable of operation at ambient temperatures as high as +131°F and of being stored at temperatures as high as +160°F. Before, during and after testing as specified in 4.9.1.2 the BTS shall meet the performance specified in SM-A-920096.

3.6.2 Altitude. The BTS shall be capable of operation at 15,000 feet altitude and of being air transported at 40,000 feet. Before, during and after testing as specified in 4.9.2 the BTS shall meet the performance specified in SM-A-920096.

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3.6.3 Humidity. The BTS shall be operable without degradation in specified performance during exposure to extreme high humidity levels. (94% RH at 86°F) and shall sustain no physical damage after prolonged exposure to extreme high humidity levels (to be determined by cycling from 70°F to 149°F at 94% RH) as encountered in tropical areas. Before, during and after testing as specified in 4.9.3 the BTS shall meet the performance specified in SM-A-920096. The BTS shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials.

3.6.4 Vibration cycling. The BTS shall be capable of operation and shall sustain no physical damage after being subjected to the vibrations normally encountered by equipment transported by rail, sea, truck or semitrailer. Before and after testing as specified in 4.9.4 the BTS shall meet the performance specified in SM-A-920096.

3.6.5 Loose cargo bounce. The BTS shall be capable of operation and shall sustain no major physical damage (cosmetic damage permitted) after being subjected to the shocks normally encountered during transportation as loose cargo. Before and after testing as specified in 4.9.5 the BTS shall meet the performance specified in SM-A-920096.

3.6.6 Bench handling. The BTS shall be capable of operation and shall sustain no physical damage after exposure to the shock intensities normally encountered during servicing. Before and after testing as specified in 4.9.6 the BTS shall meet the performance specified in SM-A-920096.

3.7 Electromagnetic interference. The equipment shall comply with the emission and susceptibility requirements of MIL-STD-461A Notice 4 (EL) as follows when tested as specified in 4.10.

CE02	CS02	RE02	RS02
	CS06	RE02.1	RS03

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a. RS03. Radiated field and modulation requirements are as follows:

Frequency	Field Intensity	Modulation
10 kHz - 1.9 MHz	1 Volt/meter	400 Hz square wave, 50% modulated
2 MHz - 75.9 MHz	20 Volt/meter	FM, ± 8 kHz deviation 1000 Hz tone
76 MHz - 399.9 MHz	20 Volt/meter	AM, 50% modulated, 1000 Hz tone
400 MHz - 1999 MHz	10 Volt/meter	Pulse, 0.1 usec width, 400 pps
2 GHz - 12.4 GHz	5 Volt/meter	Pulse, 0.1 usec width, 400 pps

b. RE02/RE02.1. The narrowband emission levels shall have a 6 dB amplitude tolerance. The military standard limit shall be relaxed 12 dB at 200 MHz.

(1) The broadband limit shall be as specified in the military standard over the 0.014 to 1000 MHz frequency range and shall be a constant 70 dBuV/m/MHz over the 1 to 20 GHz frequency range.

(2) The narrowband limit shall be as specified in the military standard over the 0.014 to 10,000 MHz frequency range and shall be a constant 60 dBuV/m over the 10 to 20 GHz frequency range.

3.8 Burn-in. Each BTS to be delivered on contract, or to be subjected to first article tests in accordance with this specification, shall be subjected to a 48 hour burn-in to limit infant mortality. The last 24 hours of the burn-in period shall be faulure free. This burn-in shall be prior to any or all tests specified herein.

3.9 Systems safety engineering.

3.9.1 Personnel hazards. Personnel hazards shall be kept to a minimum by control of configuration changes and parts selection to assure compliance with Requirement 1 of MIL-STD-454. Compliance with these requirements shall be verified through a visual inspection (see 4.5).

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3.9.2 Edge rounding, exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.03 inch (0.75 mm).

3.10 Human engineering. Configuration design criteria affecting human factors operation, maintainability and handling shall be maintained through compliance with MIL-STD-1472.

3.11 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or the system and shall conform to Requirement 7 of MIL-STD-454 (see 4.11). Individual items shall not be selected for fit or performance, however, matched pairs or sets when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating or characteristic.

3.12 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268K(EL)

General requirements for plastic material and plastic parts
Wiring and Cabling, including:
  Slack
  Protection
  Clearance
  Splicing
Connections, general
Grounding
  Wire and cable shielding
Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include packaging (see 4.4)).

b. Inspection covered by subsidiary documents (see 4.5).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.6).

(2) Quality conformance inspection of packaging (see 4.12).

4.3 Burn-in. Prior to the start of any or all of the following tests, each BTS shall be burned-in in accordance with SM-A-920182 and shall meet the requirements of that procedure and 3.8 herein.

4.4 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.4.1 First article units. The contractor shall furnish three first article units of the complete BTS.

4.4.2 First article inspection. The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified herein.

4.4.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order (see 6.5).

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract or purchase order.

ITEM	WHERE REQUIRED
Parts, materials, and processes; general	3.3
Finish	3.3.1
Marking	3.4
Safety	3.9
Human engineering	3.10

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TABLE I. First article inspection.

INSPECTIONS	REQ. PARA.	TEST PARA.	UNIT 1	UNIT 2	UNIT 3
1. Inspection covered by subsidiary documents.	3.3,3.4 3.9,3.10	4.5	Inspections to be performed on all units.		
2. Group A inspection	See Table II		Inspections to be performed on all units.		
3. Group B inspection	See Table III		Inspections to be performed on all units.		
4. Group C inspection	As follows:				
EMI	3.7	4.10		1	
Environmental					
Temperature	3.6.1	4.9.1	1		
Altitude	3.6.2	4.9.2	2		
Humidity	3.6.3	4.9.3	3		
Vibration	3.6.4	4.9.4			2
Loose cargo bounce	3.6.5	4.9.5			3
Bench handling	3.6.6	4.9.6			1

NOTES: a. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspection 4 specify the order of inspections for the indicated unit.

4.6 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.5 and 4.6.1 through 4.6.4 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.6.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using general inspection level II and the AQL's indicated in Table II.

4.6.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE II. Group A inspection.

INSPECTION	REQ. PARA.	TEST PARA.	AQL MAJOR	AQL MINOR
Visual and mechanical	3.2,3.3,3.4, 3.9,3.10,3.12	4.5, 4.7	1%	4%
Control panel	3.5.2	4.8.3	*1.5% for the group	
Status signals	3.5.3.1	4.8.4		
Forcing functions	3.5.3.2	4.8.5		
Voltage monitor	3.5.3.3	4.8.6		
Oscilloscope monitor	3.5.3.4	4.8.7		
Lamp test	3.5.3.5	4.8.8		
RF output	3.5.4.1	4.8.9		
Pulse characteristics	3.5.4.2.1, 3.5.4.2.2	4.8.10		
Con scan modulation	3.5.4.2.3	4.8.11		
Blanking output	3.5.4.4	4.8.13		
Combined RF output	3.5.5	4.8.14		
RF in	3.5.6	4.8.15		

*All electrical defects are considered major.

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4.6.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. For the Group B interchangeability requirement, test samples may be separate units whose modules and replaceable parts were randomly selected and assembled prior to Group A testing. This inspection shall conform to Table III.

4.6.2.1 Sampling for Group B inspection. Three samples of the equipment shall be selected at random from every 30 sets or fraction thereof produced. Successful testing of the first article samples shall satisfy the Group B requirement for the first 30 units produced. Failure of a Group B requirement in test fails the lot from which the Group B sample was taken.

4.6.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE III. Group B inspection.

INSPECTION	REQ. PARA.	TEST PARA.
Primary power	3.5.1	4.8.2
Sync outputs	3.5.4.3	4.8.12
Interchangeability	3.11	4.11

4.6.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.6.3.1.

4.6.3.1 Sampling for Group C inspection. Three samples of the equipment shall be selected at random from every 30 sets or fraction thereof produced. One sample shall be used for each Subgroup inspection in Table IV. Successful testing of the first article samples shall satisfy the Group C requirement for the first 30 units produced.

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4.6.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in Table IV.

4.6.3.3 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.6.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group B and Group A inspection, in that order.

TABLE IV. Group C inspection.

INSPECTION	REQ. PARA.	TEST PARA.
Subgroup I		
EMI	3.7	4.10
Subgroup II		
Temperature	3.6.1	4.9.1
Altitude	3.6.2	4.9.2
Humidity	3.6.3	4.9.3
Subgroup III		
Bench handling	3.6.6	4.9.6
Vibration	3.6.4	4.9.4
Loose cargo bounce	3.6.5	4.9.5

4.7 Visual and mechanical inspection. The equipment shall be visually and mechanically inspected to insure conformance to the construction requirements of 3.2 and the workmanship requirements of 3.12. Defects shall be classified in accordance with MIL-STD-252(EL).

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4.8 Performance tests.

4.8.1 Standard test conditions. Unless otherwise specified, the electrical tests shall be conducted under the following conditions:

a. Temperature and humidity ----------room ambient
b. Input voltage ---- 105-130 volts AC,50-60 Hz, 1Ø
c. Warm-up time ---------- 5 minutes

4.8.2 Primary power. The BTS shall meet the requirements of 3.5.1 when tested in accordance with 3.14 of SM-A-920091.

4.8.3 Control panel. The BTS shall meet the requirements of 3.5.2 when tested in accordance with 3.13 of SM-A-920091.

4.8.4 Status signals. The BTS shall meet the requirements of 3.5.3.1 when tested in accordance with 3.8 of SM-A-920091.

4.8.5 Forcing functions. The BTS shall meet the requirements of 3.5.3.2 when tested in accordance with 3.9 of SM-A-920091.

4.8.6 Voltage monitor. The BTS shall meet the requirements of 3.5.3.3 when tested in accordance with 3.11 of SM-A-920091.

4.8.7 Oscilloscope monitor. The BTS shall meet the requirements of 3.5.3.4 when tested in accordance with 3.10 of SM-A-920091.

4.8.8 Lamp test. The BTS shall meet the requirements of 3.5.2.2 & 3.5.3.5 when tested in accordance with 3.12 of SM-A-920091.

4.8.9 RF output. The BTS shall meet the requirements of 3.5.4.1 when tested in accordance with 3.1 and 3.2 of SM-A-920091.

4.8.10 Pulse characteristics. The BTS shall meet the requirements of 3.5.4.2.1 and 3.5.4.2.2 when tested in accordance with 3.3.3.5 and 3.6 of SM-A-920091.

4.8.11 Con scan modulation. The BTS shall meet the requirements of 3.5.4.2.3 when tested in accordance with 3.4 of SM-A-920091.

4.8.12 Sync outputs. The BTS shall meet the requirements of 3.5.4.3 when tested in accordance with 3.15 of SM-A-920091.

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4.8.13 Blanking output. The BTS shall meet the requirements of 3.5.4.4 when tested in accordance with 3.7 of SM-A-920091.

4.8.14 Combined RF output. The BTS shall meet the requirements of 3.5.5 when tested in accordance with 3.1.3, 3.1.4 and 3.1.7 of SM-A-920091.

4.8.15 RF in. The BTS shall meet the requirements of 3.5.6 when tested in accordance with 3.3 of SM-A-920091.

4.9 Environmental tests. All environmental test methods, conditions and procedures given below are contained in MIL-STD-810C. The equipment performance measurements shall be made under standard conditions as defined in MIL-STD-810C. Successful completion of any particular test shall be considered evidence of meeting the corresponding service condition.

4.9.1 Temperature. The equipment shall be tested according to the following test methods.

4.9.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure 1 of MIL-STD-810C and shall meet the requirements of 3.6.1.1.

Storage temperature (Step 2) ..... -70°F (-57°C)
Operating temperature (Steps 4, 5) +32°F (0°C)
Make measurements at Steps 1, 5, 7.

4.9.1.2 High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure II of MIL-STD-810C and shall meet the requirements of 3.6.1.2.

Storage temperature (Steps 2,3,5) +120°F(+49°C)
Storage temperature (Step 4) +160°F(+71°C)
Operating temperature(Steps 7,8) +131°F(+55°C)
Make measurements at Steps 1, 8, 10.

4.9.2 Altitude. The equipment shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810C. A non-operating test shall be performed for 1 hour at 40,000 feet. The equipment shall meet the requirements of 3.6.2 when tested at Steps 1, 2 and 4.

Operating altitude ..... 15,000 feet
Non-operating altitude ..... 40,000 feet

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4.9.3 Humidity. The equipment shall be tested in accordance with Method 507.1, Procedure II of MIL-STD-810C. The equipment shall be operated during the last 5 hours of the last cycle of Step 6, and shall meet the requirements of 3.6.3 when tested in Steps 4, 6 (last cycle only) and 8.

4.9.4 Vibration cycling. The equipment in its combination case shall be tested in accordance with Method 514.2, Procedure X of MIL-STD-810C, and shall meet the requirements of 3.6.4.

Equipment category	g
Applicable table	514.2 - VII
Applicable figure	514.2 - 7
Test curve	AW

Make measurements before and after only. The frequency of applied vibration shall be swept logarithmically over the specified range.

4.9.5 Loose cargo bounce. The equipment shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810C, and shall meet the requirements of 3.6.5.

4.9.6 Bench handling. The equipment shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810C, and shall meet the requirements of 3.6.6.

4.10 Electromagnetic interference. The equipment shall be tested in accordance with the test setups and measurement techniques of MIL-STD-462, Notice 3 as implemented by a contractor prepared Government approved EMI test plan and shall meet the requirements of 3.7. These measurements shall have a $\pm$ 3% frequency tolerance.

4.11 Interchangeability. The equipment shall be inspected to insure the physical and functional interchangeability requirements of 3.11.

4.11.1 Electrical interchangeability. All like modules and replaceable parts shall be tested for functional interchangeability by assembly of randomly selected parts, when inspected in accordance with 3.16 of SM-A-920091. The assembly shall be tested to and shall meet Group A acceptance requirements. Electrical retuning or realignment of modules during production assembly may be expected and are permissible. Random selection for this assembly shall be supervised by the Government.

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4.11.2 Mechanical interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11, when inspected in accordance with 4.7 of SM-A-920091. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.12 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each BTS shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each BTS shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Technical literature. Each item of technical literature shall be individually unit packed method IC-1.

5.1.1.4.2 BTS. Each set shall be individually unit packed method III as follows: Secure each major component in combination case. Place the ancillary equipment in the cover of the case. Position the cover and secure with latches provided. Each combination case shall be cushioned on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety TW (Triple-Wall), designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic modling material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be enclosed within a close-fitting fiberboard box conforming to PPP-B-636, VIIc. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservation application and unit packing shall be as specified in 5.1.1.

5.2 Packing. Packing shall be level A or B as specified (see 6.2b). Shipping containers shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Each BTS unit packed as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. 3 x 4 inch skids laid flat shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

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5.2.2 Level B. Each BTS, unit packed as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. 3 x 4 inch skids, laid flat shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will addquately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.3 Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1. Intended use. The BTS is the test equipment which is used to test the operation of the CMS at the direct support level of maintenance (AVIM). The CMS is used aboard U.S. Army helicopters as protection against radar directed weapon systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When rough handling is required.

d. First article inspection.

Three complete systems are required so that lengthy environmental tests can be conducted on sample units.

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6.2 Ordering data (cont'd).

e. Place of final inspection.

f. Technical literature required.

g. Quantity of spare parts required.

6.3 Definitions.

6.3.1 Temperature stabilization. Temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag does not change more than 2.0°C (3.6°F) per hour.

6.3.2 Physical damage. Physical damage is defined as corrosion, rust, blistering, swelling or other deterioration of parts and materials.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Data requirements. Deliverable data required by this specification is cited in 4.4.3 entitled first article data.

6.6 First article. First articles are required for inspection and approval (see 3.1, 4.4 and 6.2).

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging, material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 CMS primary power. The BTS shall be capable of supplying 980 watts at 28 Vdc to the CMS, as obtained from an external source.

6.9 BTS RF input signal characteristics. The BTS is capable of operating with input signals having the following characteristics.

Frequency:	A to E +0.5 GHz
Power:	400 mW peak
Duty cycle:	6%
Pulse width:	As low as 0.2 microsecond
Impedance:	50 ohms, nominal

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6.10 PRI modulation. PRI modulation for threat A-W, threat A-F, and threat B modes resides in digital format and program changes are accomplished by replacing a minimum number of plug-in type integrated circuits (ROMS). A program capability of 512 words, each word consisting of 20 bits, is provided. The program data is generated directly from data measured on the threat radar system.

6.11 Weight. The weight of the BTS is 230 pounds, exclusive of the GFE.

6.12 Size. The size of the BTS is 36 inches high, 20 inches wide and 26.5 inches deep.

6.13 Cooling. Two cooling fans are provided in the BTS. One is for cooling of the BTS electronics while the other provides cooling air to LRU-1 of the CMS.

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Specification MIL-T-49364(ER) 21 Mar 80

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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